

Lack of Awareness among Future Medical Professionals about the Risk of Consuming Hidden Phosphate-Containing Processed Food and Drinks



Phosphate toxicity is an important determinant of mortality in patients with chronic kidney disease (CKD), particularly those undergoing hemodialysis treatments. CKD patients are advised to take a low phosphate-containing diet, and are additionally prescribed with phosphate-lowering drugs. Since these patients usually seek guidance from their physicians and nurses for their dietary options, we conducted a survey to determine the levels of awareness regarding the high phosphate content in commercially processed food and drinks among medical and nursing students at the Hirosaki University School of Medicine in Japan. For this survey, 190 medical and nursing students (average age 21.7 ± 3 years) were randomly selected, and provided with a list of questions aimed at evaluating their awareness of food and drinks containing artificially added phosphate ingredients. While 98.9% of these students were aware of the presence of sugar in commercially available soda drinks, only 6.9% were aware of the presence of phosphate (phosphoric acid). Similarly, only 11.6% of these students were aware of the presence of phosphate in commercially processed food, such as hamburgers and pizza. Moreover, around two thirds of the surveyed students (67.7%) were unaware of the harmful effects of unrestricted consumption of phosphate-containing food and drinks. About 28% of the surveyed students consume such “fast food” once a week, while 40% drink at least 1–5 cans of soda drinks/week. After realizing the potential long-term risks of consuming excessive phosphate-containing food and drinks, 40.5% of the survey participants considered reducing their phosphate intake by minimizing the consumption of commercially processed “fast food” items and soda drinks. Moreover, another 48.4% of students showed interest in obtaining more information on the negative health effects of consuming excessive amounts of phosphate. This survey emphasizes the need for educational initiative to raise awareness of the health risks posed by excessive consumption of phosphate additives.